

Input-Output map derivation from Power Flow Manifold:

Starting from the power flow manifold of Bolognani and Dörfler to $A_{\text{sh}}x \leq b_{\text{sh}}$ form for GNE shared constraints.

Pipeline:

1. from the tangent linear manifold (Paper, Eq. 1–5) $A_{x^*}(x - x^*) = d_{x^*}$ and the augmented bus-model manifold $C_{x^*}(x - x^*) = d_{x^*}$;
2. partitioning of x into unknowns $\mathcal{U}$ and knowns (depending on bus type i.e. generation (PV), load (PQ), slack), $\mathcal{K}$, worked out explicitly for a PQ node;
3. the resulting input–output map $y = Su + m$, ($y = Mx + m$),
4. the voltage-band constraint $\underline{v} \leq y \leq \bar{v}$ rewritten and stacked as a one-sided shared constraint $A_{\text{sh}}u \leq b_{\text{sh}}$, the form required by the GNE/DyNECT pipeline.

1. Starting point: the manifold

Power flow equations written implicitly as $F(x) = 0_{2n}$, $x = (v, \theta, p, q)$. At a linearization point x^* , Proposition 1 gives the tangent plane

$$A_{x^*}(x - x^*) = 0_{2n}, \quad A_{x^*} = [A_{v\theta} \quad -I_{2n}].$$

Note: this is the *deviation* form. In absolute coordinates it would be $A_{x^*}x = d_{x^*}$ with $d_{x^*} := A_{x^*}x^*$. I work in deviations throughout, and this constant d_{x^*} resurfaces later as the offset m .

The bus models $g_h(x_h) = 0_2$ (PQ, PV, slack) are stacked into $G(x) = 0_{2n}$, with tangent $C_{x^*}(x - x^*) = d_{x^*}$, $C_{x^*} = \partial G / \partial x|_{x^*}$. Combined with A_{x^*} this gives the $4n \times 4n$ system (Eq. 13 in the paper) whose solution is the first-order approximation $\hat{x}$.

2. Splitting effects

Each bus has 4 variables $(v_h, \theta_h, p_h, q_h)$. The bus model fixes 2 of them (setpoints) and leaves 2 free (solved from A_{x^*}). So:

$$\mathcal{K} = \{\text{fixed by bus model}\}, \quad \mathcal{U} = \{\text{free, solved from network eqs.}\}$$

For a **PQ bus**, the model is $g_h = (p_h - P_h, q_h - Q_h) = 0$, so:

$$\mathcal{K}_h = \{p_h, q_h\}, \quad \mathcal{U}_h = \{v_h, \theta_h\}.$$

Input/output definitions for this PQ bus:

$$u_h = \begin{bmatrix} \Delta p_h \\ \Delta q_h \end{bmatrix}, \quad y_h = \begin{bmatrix} \Delta v_h \\ \Delta \theta_h \end{bmatrix}, \quad u_h^* = \begin{bmatrix} p_h^* \\ q_h^* \end{bmatrix}, \quad y_h^* = \begin{bmatrix} v_h^* \\ \theta_h^* \end{bmatrix}.$$

u^*, y^* are just the values of (p, q, v, θ) at the linearization point.

3. Explicit Derivation

Step 1 — Nonlinear power flow manifold

Let $x = (v, \theta, p, q) \in \mathbb{R}^{4n}$ collect voltage magnitudes, angles, active and reactive injections for an n -bus network. The AC power flow equations define the manifold

$$\mathcal{M} = \{x \mid F(x) = 0_{2n}\}.$$

Assume $x^* \in \mathcal{M}$ is a known operating point.

Step 2 — Tangent linear approximation of $\mathcal{M}$

Around x^* , the Bolognani–Dörfler tangent plane is

$$A_{x^*}(x - x^*) = 0_{2n}, \quad A_{x^*} = \begin{bmatrix} A_{v\theta} & -I_{2n} \end{bmatrix},$$

which, writing $\Delta z = (\Delta v, \Delta \theta)$ and $\Delta s = (\Delta p, \Delta q)$, gives

$$A_{v\theta} \Delta z = \Delta s.$$

This is the centered (deviation) form. In absolute coordinates: $A_{x^*}x = d_{x^*}$, $d_{x^*} := A_{x^*}x^*$. Working in deviations, d_{x^*} does not appear explicitly; it reappears at the end as the offset m .

Step 3 — Bus-model tangent and combined system

Each bus h carries a bus model $g_h(x_h) = 0_2$ that fixes two of the four local variables (PQ: p_h, q_h ; PV: p_h, v_h ; slack: v_h, θ_h). Stacking all bus models gives $G(x) = 0_{2n}$, with tangent

$$C_{x^*}(x - x^*) = d_{x^*}, \quad C_{x^*} := \left. \frac{\partial G}{\partial x} \right|_{x^*}, \quad d_{x^*} := -G(x^*).$$

Stacking the manifold tangent and the bus-model tangent gives the combined first-order system whose solution $\hat{x}$ approximates the true power flow:

$$\begin{bmatrix} A_{x^*} \\ C_{x^*} \end{bmatrix} (\hat{x} - x^*) = \begin{bmatrix} 0_{2n} \\ d_{x^*} \end{bmatrix}.$$

C_{x^*} is what formally justifies classifying exactly two variables per bus as known (fixed by the bus model) and two as unknown (determined by A_{x^*}).

Step 4 — Slack bus handling

Let bus 0 be the slack bus. Its voltage magnitude and angle are fixed references: $\Delta v_0 = 0$, $\Delta \theta_0 = 0$. The slack bus is excluded from the input-output map entirely:

- Its columns $(\Delta v_0, \Delta \theta_0)$ are removed from $A_{v\theta}$ (they are zero).
- Its power-balance rows are not part of the reduced map; they are retained separately to recover slack injections $\Delta p_0, \Delta q_0$ in Step 12.

This leaves a reduced system over the $n - 1$ non-slack buses:

$$A^{\text{red}} \begin{pmatrix} \Delta v_{\setminus 0} \\ \Delta \theta_{\setminus 0} \\ \Delta p_{\setminus 0} \\ \Delta q_{\setminus 0} \end{pmatrix} = 0_{2n-2}, \quad A^{\text{red}} \in \mathbb{R}^{(2n-2) \times (4n-4)}.$$

Step 5 — Per-bus unknown / known split

The bus model g_h fixes two variables per non-slack bus and leaves two free. Classify accordingly:

Bus type	Unknown $\mathcal{U}$	Known / input $\mathcal{K}$
PQ	$\Delta v_i, \Delta \theta_i$	$\Delta p_i, \Delta q_i$
PV	$\Delta \theta_i, \Delta q_i$	$\Delta p_i, \Delta v_i$

Stack all unknowns into $\Delta x_{\mathcal{U}} \in \mathbb{R}^{2(n-1)}$ and all knowns into $\Delta x_{\mathcal{K}} \in \mathbb{R}^{2(n-1)}$:

$$\Delta x_{\mathcal{U}} = \begin{pmatrix} \Delta v_{PQ} \\ \Delta \theta_{PQ} \\ \Delta \theta_{PV} \\ \Delta q_{PV} \end{pmatrix}, \quad \Delta x_{\mathcal{K}} = \begin{pmatrix} \Delta p_{PQ} \\ \Delta q_{PQ} \\ \Delta p_{PV} \\ \Delta v_{PV} \end{pmatrix}.$$

Step 6 — Reorder the reduced system

Reorder the columns of A^{red} into the $(\mathcal{U}, \mathcal{K})$ ordering:

$$A^{\text{red}} \begin{pmatrix} \Delta x_{\mathcal{U}} \\ \Delta x_{\mathcal{K}} \end{pmatrix} = 0_{2n-2} \implies A_{\mathcal{U}} \Delta x_{\mathcal{U}} + A_{\mathcal{K}} \Delta x_{\mathcal{K}} = 0_{2n-2},$$

where $A_{\mathcal{U}}, A_{\mathcal{K}} \in \mathbb{R}^{(2n-2) \times (2n-2)}$ are both square by construction. The critical condition for all that follows is

$$\det(A_{\mathcal{U}}) \neq 0.$$

Step 7 — Solve for the unknowns

Rearrange:

$$A_{\mathcal{U}} \Delta x_{\mathcal{U}} = -A_{\mathcal{K}} \Delta x_{\mathcal{K}}.$$

Since $A_{\mathcal{U}}$ is nonsingular:

$$\boxed{\Delta x_{\mathcal{U}} = -A_{\mathcal{U}}^{-1} A_{\mathcal{K}} \Delta x_{\mathcal{K}}.}$$

Step 8 — Define the input

$$u := \Delta x_{\mathcal{K}} = \begin{pmatrix} \Delta p_{PQ} \\ \Delta q_{PQ} \\ \Delta p_{PV} \\ \Delta v_{PV} \end{pmatrix}.$$

Step 9 — Define monitored outputs

A monitored output may live in $\Delta x_{\mathcal{U}}$ (e.g. PQ-bus voltages, non-slack angles, PV-bus reactive outputs) or in $\Delta x_{\mathcal{K}}$ (e.g. PV-bus voltage magnitudes, which are known inputs). Write

$$\Delta y = C_{\mathcal{U}} \Delta x_{\mathcal{U}} + C_{\mathcal{K}} \Delta x_{\mathcal{K}},$$

where $C_{\mathcal{U}}$ and $C_{\mathcal{K}}$ are row-selection matrices. Each output row is nonzero in exactly one of $C_{\mathcal{U}}, C_{\mathcal{K}}$.

Note: $C_{\mathcal{K}} \neq 0$ is required whenever a monitored output is a PV-bus voltage magnitude, since $\Delta v_{PV} \in \Delta x_{\mathcal{K}}$.

Step 10 — Substitute and form S

Substitute Step 7 into Step 9:

$$\Delta y = C_{\mathcal{U}}(-A_{\mathcal{U}}^{-1}A_{\mathcal{K}}\Delta x_{\mathcal{K}}) + C_{\mathcal{K}}\Delta x_{\mathcal{K}} = (C_{\mathcal{K}} - C_{\mathcal{U}}A_{\mathcal{U}}^{-1}A_{\mathcal{K}})\Delta x_{\mathcal{K}}.$$

Since $u = \Delta x_{\mathcal{K}}$:

$$\Delta y = Su, \quad S := C_{\mathcal{K}} - C_{\mathcal{U}}A_{\mathcal{U}}^{-1}A_{\mathcal{K}}.$$

Step 11 — Absolute-coordinate affine map

Let u^* and y^* be the values of the input and output at the linearization point. Then $\Delta y = y - y^*$ and $u = u^{abs} - u^*$, so

$$y - y^* = S(u^{abs} - u^*) \implies y = Su^{abs} + (y^* - Su^*).$$

Define the affine offset

$$m := y^* - Su^*.$$

Relabelling $u^{abs} \rightarrow u$ (so u denotes absolute injections), the map is

$$y = Su + m, \quad m = y^* - Su^*.$$

The offset m absorbs both the operating-point shift and the constant d_{x^*} from Step 3, consistent with the absolute-form $A_{x^*}x = d_{x^*}$ noted in Step 2.

Step 12 — PQ-only sanity check

If all non-slack buses are PQ, then $A_{\mathcal{K}} = -I$ (from $A_{v\theta}\Delta z - \Delta s = 0$) and $C_{\mathcal{K}} = 0$ (all outputs are voltage states). Then

$$S = C_{\mathcal{K}} - C_{\mathcal{U}}A_{\mathcal{U}}^{-1}A_{\mathcal{K}} = 0 - C_{\mathcal{U}}A_{\mathcal{U}}^{-1}(-I) = C_{\mathcal{U}}A_{\mathcal{U}}^{-1},$$

recovering the standard all-PQ sensitivity map.

Step 13 — Slack injection post-processing

The slack-bus power-balance rows dropped in Step 4, reordered consistently with $(\Delta x_{\mathcal{U}}, \Delta x_{\mathcal{K}})$, give blocks $A_{0,\mathcal{U}}$ and $A_{0,\mathcal{K}}$. Once $\Delta x_{\mathcal{U}}$ and $\Delta x_{\mathcal{K}}$ are known:

$$\begin{pmatrix} \Delta p_0 \\ \Delta q_0 \end{pmatrix} = A_{0,\mathcal{U}}\Delta x_{\mathcal{U}} + A_{0,\mathcal{K}}\Delta x_{\mathcal{K}}.$$

Substituting $\Delta x_{\mathcal{U}} = -A_{\mathcal{U}}^{-1}A_{\mathcal{K}}u$ gives an affine map from u to slack injections as well.

Final result

The reduced tangent manifold induces a local affine input-output map from controllable injections and PV voltage setpoints to monitored electrical variables, provided $A_{\mathcal{U}}$ is nonsingular:

$$y = Su + m, \quad S = C_{\mathcal{K}} - C_{\mathcal{U}}A_{\mathcal{U}}^{-1}A_{\mathcal{K}}, \quad m = y^* - Su^*.$$

4. Shared-constraint form

We want $\underline{v} \leq y \leq \bar{v}$. Substitute $y = Su + m$:

$$\underline{v} \leq Su + m \leq \bar{v}.$$

Subtracting m :

$$\underline{v} - m \leq Su \leq \bar{v} - m.$$

Splitting into two one-sided inequalities:

$$Su \leq \bar{v} - m \quad \text{and} \quad -Su \leq -\underline{v} + m.$$

(The second one comes from $\underline{v} - m \leq Su \Rightarrow -Su \leq -(\underline{v} - m) = -\underline{v} + m$.)

Stacking both on top of each other:

$$\begin{bmatrix} S \\ -S \end{bmatrix} u \leq \begin{bmatrix} \bar{v} - m \\ -\underline{v} + m \end{bmatrix}.$$

This is the target shared-constraint form

$$A_{\text{sh}} u \leq b_{\text{sh}}, \quad A_{\text{sh}} = \begin{bmatrix} S \\ -S \end{bmatrix}, \quad b_{\text{sh}} = \begin{bmatrix} \bar{v} - m \\ -\underline{v} + m \end{bmatrix}.$$